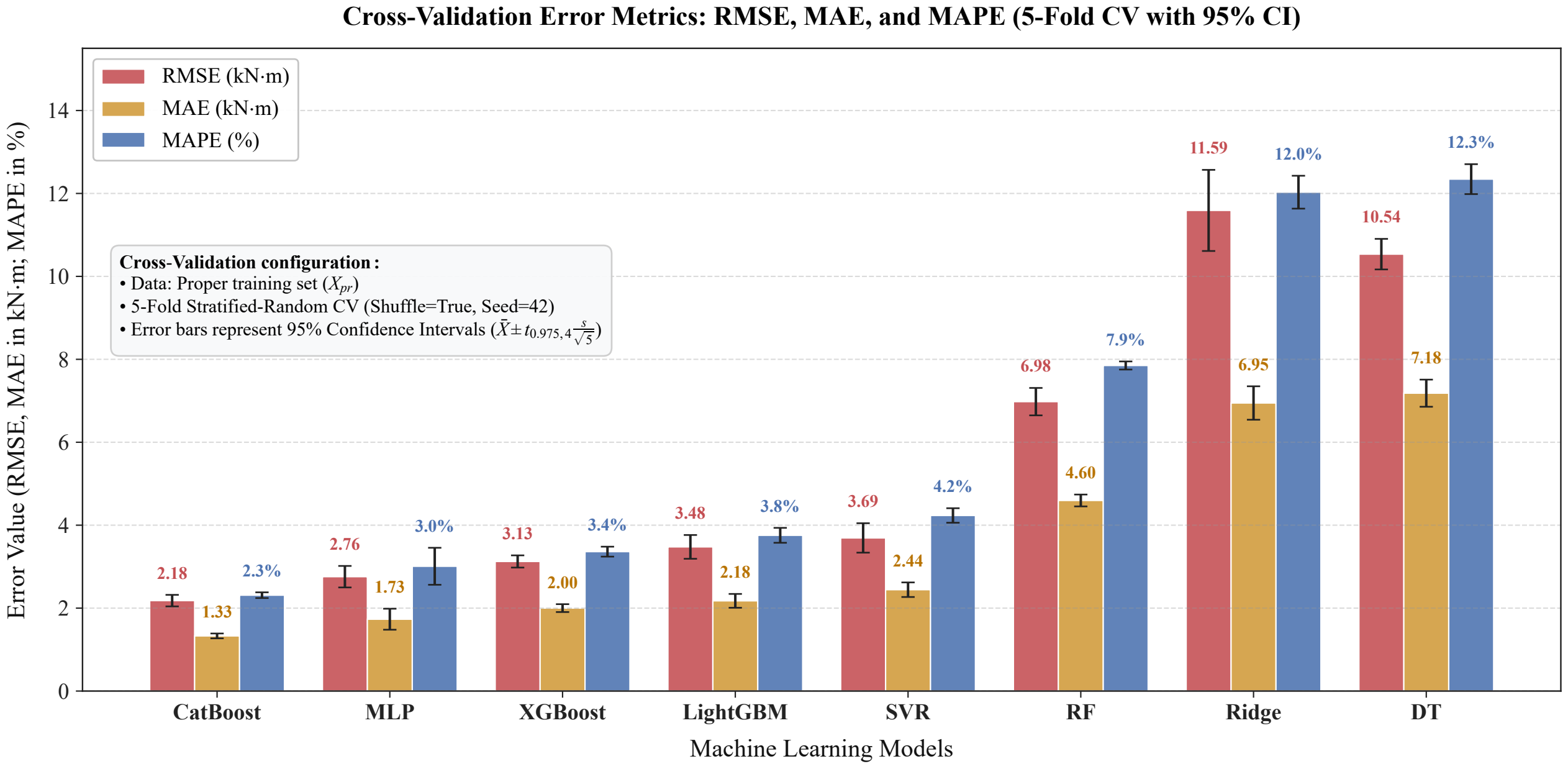

Cross-Validation Error Metrics: RMSE, MAE, and MAPE (5-Fold CV with 95% CI)
RMSE (kN·m)
MAE (kN·m)
MAPE (%)
Cross-Validation configuration :
• Data: Proper training set (X_{pr})
• 5-Fold Stratified-Random CV (Shuffle=True, Seed=42)
• Error bars represent 95% Confidence Intervals ($\bar{X} \pm t_{0.975,4}\frac{s}{\sqrt{5}}$)
Error Value (RMSE, MAE in kN·m; MAPE in %)
0
2
4
6
8
10
12
14
2.18
1.33
2.3%
2.76
1.73
3.0%
3.13
2.00
3.4%
3.48
2.18
3.8%
3.69
2.44
4.2%
6.98
4.60
7.9%
11.59
6.95
12.0%
10.54
7.18
12.3%
CatBoost
MLP
XGBoost
LightGBM
SVR
RF
Ridge
DT
Machine Learning Models